

POORNIMA KOSARAJU

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SUMMARY

Highly skilled and motivated VLSI Engineer with a solid academic foundation in electrical engineering and a keen focus on integrated circuit design and verification. Proficient in industry-standard EDA tools and languages, including Verilog, VHDL, and SystemVerilog. Demonstrates strong analytical abilities and practical experience through academic projects and internships.

EDUCATION

New Jersey Institute of Technology

Newark, NJ

Master of Science (M.S.) in Electrical Engineering | GPA: 3.6

August 2023 – May 2024

Relevant Coursework: Circuit Design (VLSI), VHDL, Embedded Systems, Computer Architecture, High performance Digital Signal Processing Laboratory.

Shri Vishnu Engineering College for Women

Bhimavaram, India

Bachelor of Technology (B. TECH) | ECE | GPA: 3.7

August 2019 - May 2023

Relevant Coursework: Digital Logic Design, Verilog, Microprocessors & Controllers, Signals & Systems, Python, C++

EXPERIENCE.

RTL Design Intern

Hyderabad, India

VEDAIIIT

January 2022 – July 2022

- Contributed to the creation of blueprints for tiny electronic components using a 28nm CMOS process.
- Achieved 95% functional coverage, meaning the designs worked as intended in most scenarios.
- Assisted in improving the efficiency and power consumption of the circuits.
- Contributed to a 10% decrease in energy usage through synthesis and timing analysis.

Embedded Systems Project

Bhimavaram, India

Shri Vishnu Engineering College for Women

September 2022 – January 2023

- Created the underlying code (firmware) for a system using C and C++ languages, ensuring it could handle tasks in real-time.
- Created a blueprint for the embedded system using ARM microcontrollers, the central processing units for these devices.
- Tested the system through simulations and physical prototypes to ensure it worked as expected.

TECHNICAL PROJECTS

- FIFO Design Using Verilog:** The FIFO module, implemented in Verilog, was designed to handle asynchronous data transfers efficiently. The module was thoroughly tested to verify its functionality under various conditions, ensuring its suitability for real-world applications.
- 16-Bit Serial Parallel Multiplier:** The 16-bit serial-parallel multiplier, designed using Mentor Graphics tools, was optimized for both speed and area efficiency. It was designed to perform calculations quickly while also taking up minimal space on the integrated circuit.
- Energy-Aware Hardware Implementation of Network Coding:** The network coding hardware was implemented using a hardware description language, focusing on energy efficiency. This involved optimizing the design to reduce power consumption without sacrificing performance. The implementation incorporated clock gating technique to control the power supply to different parts of the circuit based on their activity.

TECHNICAL SKILLS

Programming/ Scripting Languages: Python, Perl, Cshell

Hardware Description Languages & Design Tools: Verilog, HDL, SystemVerilog, ModelSim, Cadence Virtuoso, Mentor Graphics, MATLAB, XilinxISE, LINUX, HSPICE

Design Methodologies: Synthesis, Static Timing Analysis, RTL Design, Low Power Design, Signal processing

Fabrication Processes: CMOS, FinFET